

Lab 2 72418 :- BOUNCING BALL BY FRAME ANIMATION IN MACROMEDIA FLASH.

Introduction :-

Macromedia flash is a powerful tool for creating computer-based animation. This lab demonstrate the process of animating a ball bouncing down a staircase using keyframes. This animation is created manually by adjusting the position of the ball frame by frame.

Objective :-

- To understand key-frame based animation in macromedia flash.
- To create realistic bounding ball effect using keyframes.
- To manually adjust object positions for smooth animations.
- To learn about frame rates and their impact on animation smoothness.

Theory of computer - Based Animation

Computer - Based Animation is the process of creating moving images using digital tools and software. It involves generating sequence of images known as frames, that when played in succession, create the illusion of motion.

Key concept include:-

- Keyframes :- Defines the start and end of motion. These are the frames where significant changes occur in an animation.
- Frame rate :- Measured in frames per second (FPS), it determines animation smoothness. A higher FPS result in a smoother animation.
- Manual frame-by-frame Animation :- Unlike tweening, where movement is generated automatically, manual animation requires adjusting each frame individually.

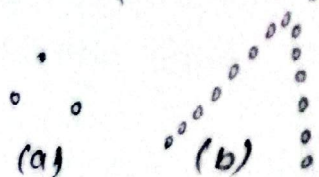


Fig. Linear interpolation of motion of ball

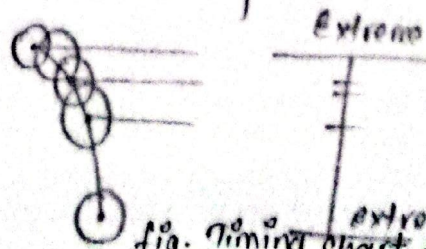


Fig. Timing chart for ball bounce

Basic steps for computer-based Animation

Input process

At the initial stage of animation, graphical elements (balls & statement) are drawn & placed on separate layers. Keyframes are inserted at points where the ball changes direction. Each keyframes represent a major position change.

Composition stage

The foreground (ball) and background (staircase) are combined. The ball's position is adjusted frame by frame to simulate bouncing. careful alignment is required to ensure the motion appears natural.

In-Between process

The animation of movement from one position to another requires intermediate frames between keyframes. The manual process ensure a smooth transition by adjusting the ball's position slightly in each frames.

Adjusting motion & speed

By modifying the distance the ball moves between frames, different speeds of motion can be simulated. For example, larger gaps between frames can indicate faster motion, while smaller gaps create a slower effect.

Practical Implementation : Bouncing Ball Animation

Steps to create Animation

- Create a new flash Document
- Set the stage size to 550 to 400 pixels.
- Set frames rate to 1 fps for smooth movement
- Draw the objects
 - Use the shape tools to create a staircase
 - Use the circle tool to create a ball

5. Insert Keyframes
 - Right click on the timeline and insert keyframes at position where the ball changes direction.
 - each keyframes makes a movement when the ball bounces.
4. manually adjust object position
 - move the ball in each keyframes to simulate bouncing down the stairs.
 - Ensure a gradual decrease in bounce height create a realistic effect.
5. Test and Export
 - Use the 'control > play' option to preview the animation.
 - Save the export the animation as a swf file.

Analysis and Discussion

The bouncing ball animation shows how important keyframing is in comput. animation. Some challenges faced during the process include.

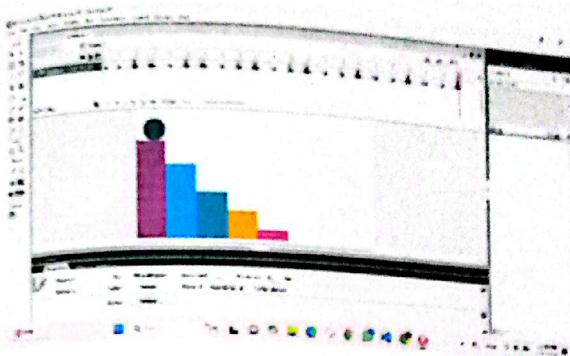
- making smooth transition between keyframes.
- Adjusting timing to make the motion look real.
- controlling the frame rate for smooth movement.
- Gradually reducing bounce height to create a natural effect.
- Using easing technique (ease-in, ease out) to make the movement look more natural.
- Keeping the ball on a consistent path for a smooth animation.

Conclusion :-

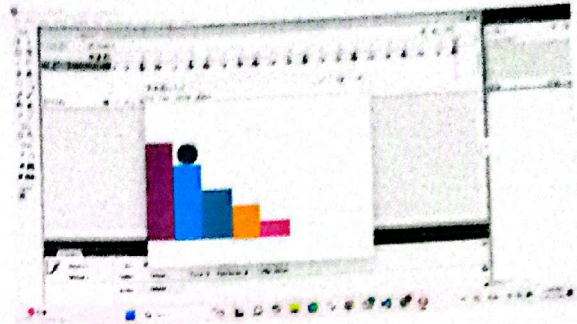
This activity helped demonstrate how to create a bouncing ball animation using macromedia flash. Important animation concepts like keyframing, frame rate control, & manual adjustment were used to make the motion smooth and natural. Learning these basic is useful for understanding animation. and preparing for more advanced motion graphics.

OUTPUT:

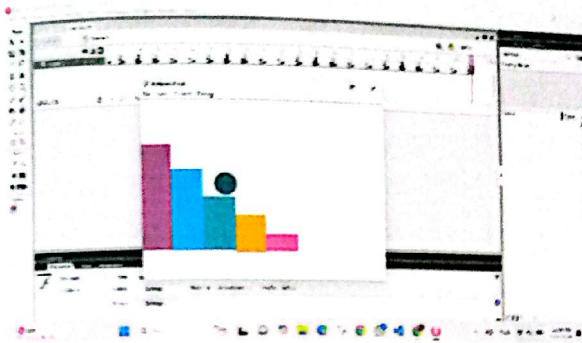
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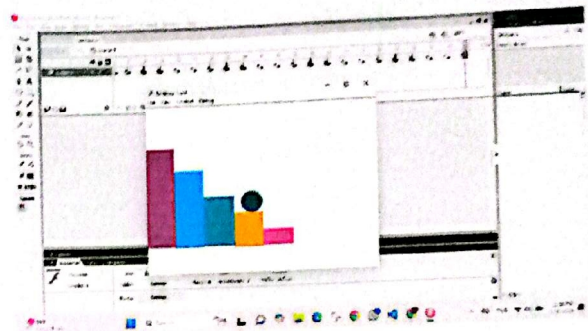
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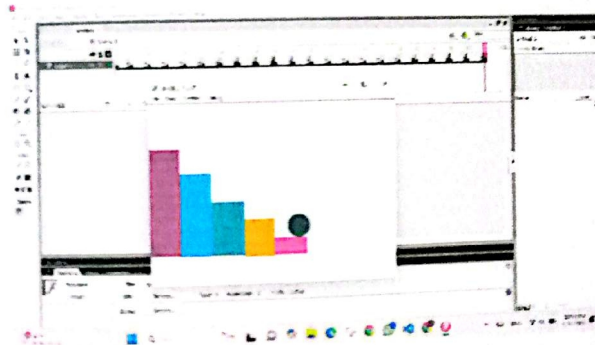
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